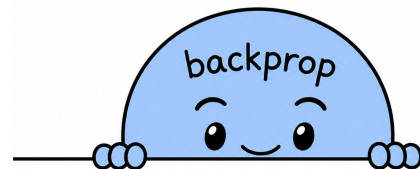


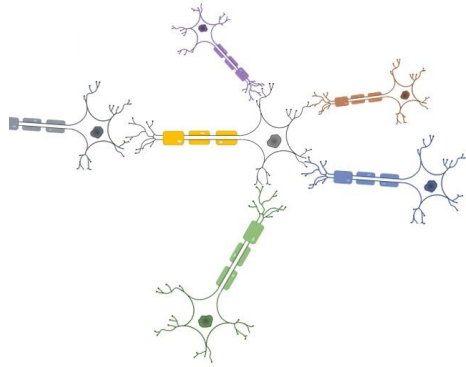
The more the merrier?

Switching between different learning rules



Team: Het Thakkar, Aayushi Vishnoi, Mauricio A. Diaz, Shekhar Nath Prakas, Bhaskar Roy, Nalin Dhiman, Yuxin Feng, Su Zhang, Vikramaditya Bisani

Problem Statement



Biological Brains

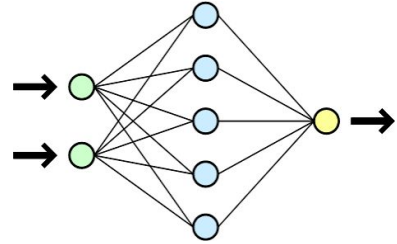
Hebbian

STDP

Neuromodulatory

Structural

Multiple plasticity mechanisms working together,
and across timescales.



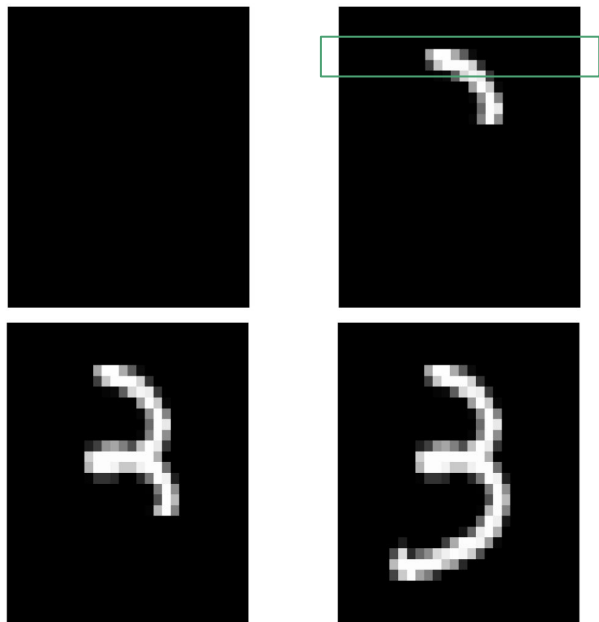
Artificial Neural Network

Backpropagation

Single rule throughout training

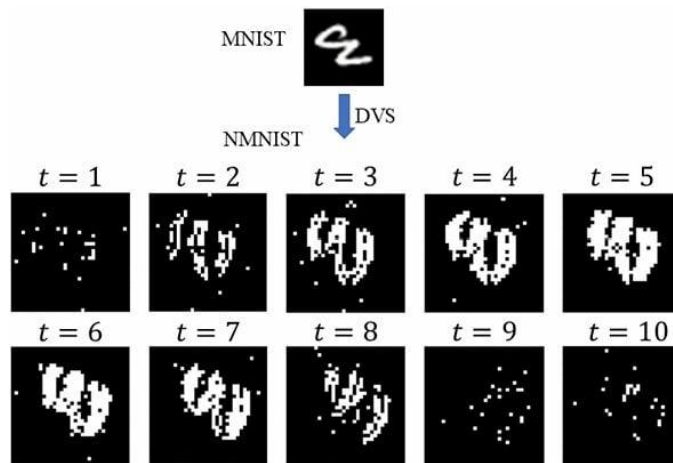
Can combining or switching between multiple learning rules produce more efficient and biologically plausible learning, rather than using a single rule?

Dataset: Temporal Variations of MNIST



Sequential MNIST (sMNIST)

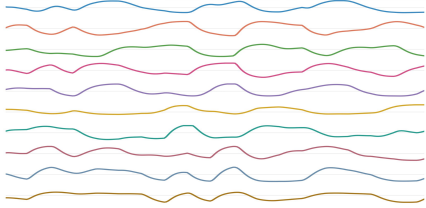
VS



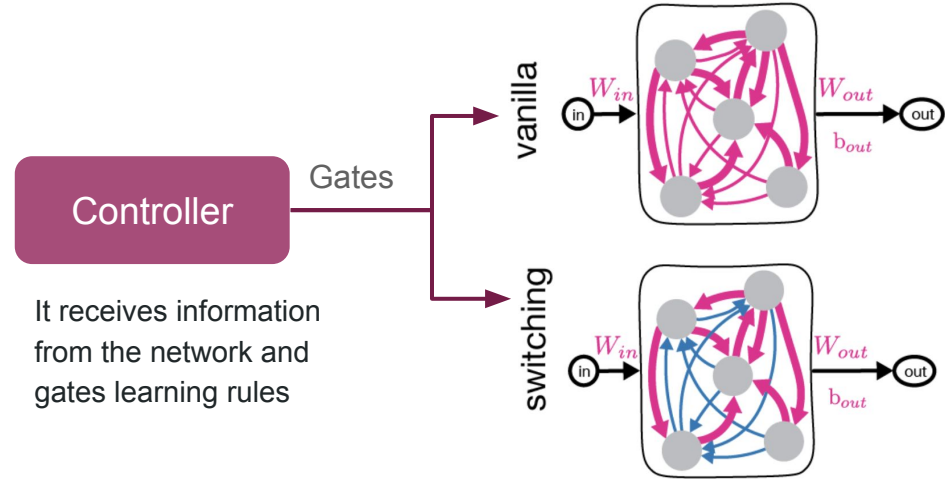
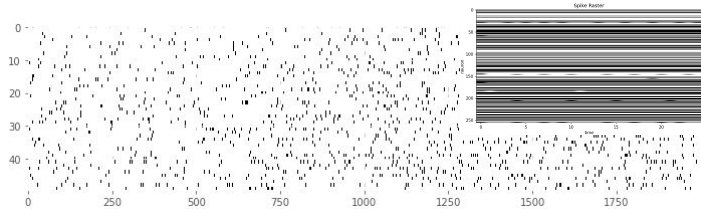
Neuromorphic MNIST (N-MNIST)

Methods

Rate-based Neural Networks (RNN)



Spiking Neural Networks (SNN)

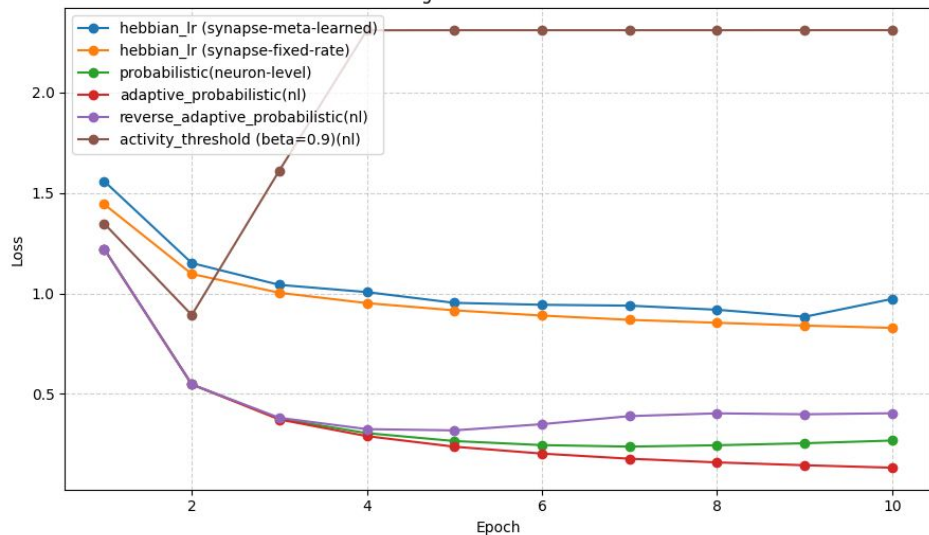


Based on framework developed by Turcu et al. (2026).

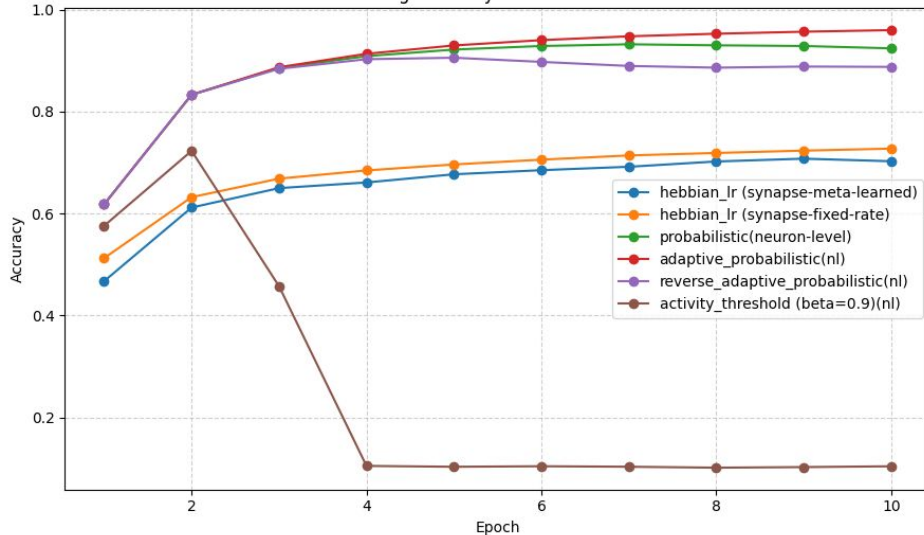
Learning using switching synaptic plasticity rules (Turcu et al., 2026)

Synaptic and neural activity switching in RNNs

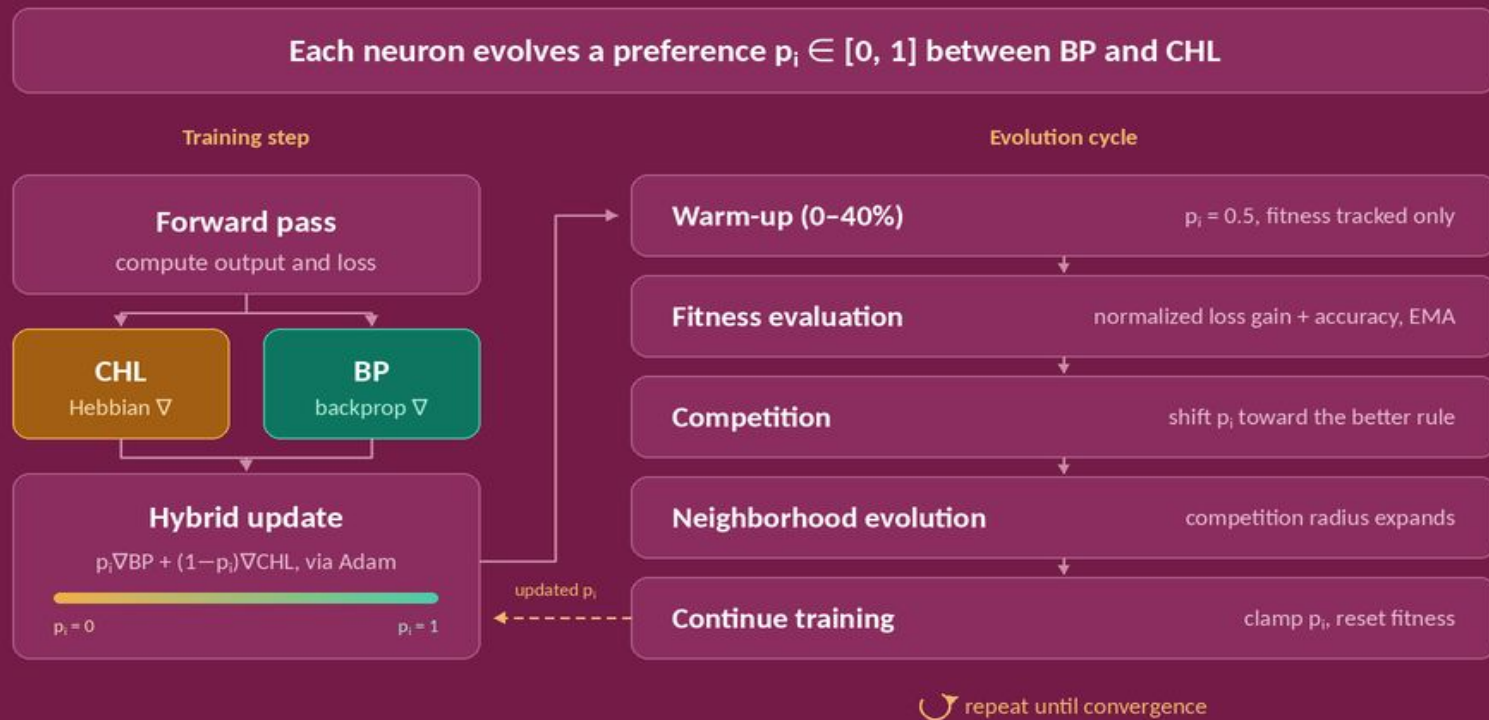
Training loss across controllers



Training accuracy across controllers



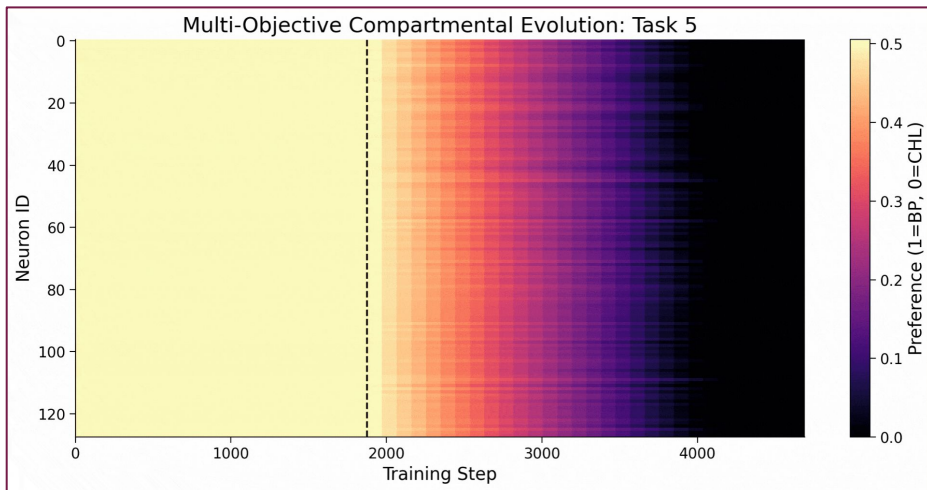
Can a neural network gradually evolve toward the locally fitter learning rule through biologically inspired competition between neighboring neurons?



Key Experimental Findings

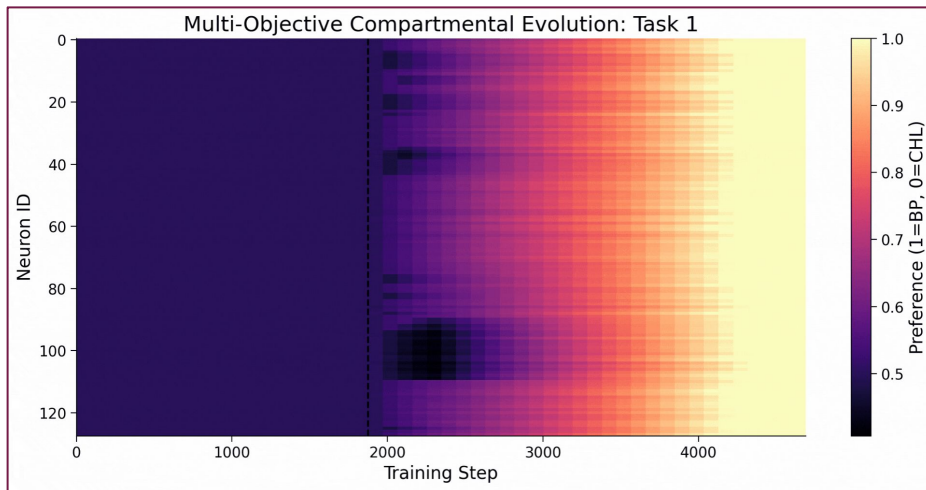
It was found that BP consistently excelled at structured geometric transformations, whereas CHL performed better under information-loss and noisy conditions, suggesting complementary inductive biases.

Tasks where Neurons Chose CHL



- Blur Augmentation (Task 5)
- Random Erasing Augmentation (Task 6)
- Mixed Augmentation (Task 8)

Tasks where Neurons Chose BP



- Normal Augmentation (Task 1)
- Affine Augmentation (Task 2)
- Elastic Augmentation (Task 3)
- Perspective Augmentation (Task 4)
- Random Transform Augmentation (Task 7)

Switching learning rules in an SNN

Main question

Can only a selected part of the recurrent network receive task-directed credit, while the rest learns using a simpler local rule?

$$\Delta W_{ji} = G_j \Delta W_{ji}^C + (1 - G_j) \Delta W_{ji}^O$$

Local credit: $G_j = 1$

$$\Delta W_{ji}^C = -\eta_C \sum_t L_j(t) e_{ji}(t)$$

- $e_{ji}(t)$: fading memory of recent synaptic influence
- $L_j(t)$: task-error signal sent to neuron j
- task-directed, eligibility-based update

Oja learning: $G_j = 0$

$$\Delta W_{ji}^O = \eta_O \sum_t r_j(t) [x_i(t) - r_j(t) W_{ji}]$$

- uses only local pre/post activity
- strengthens correlated activity
- normalization limits weight growth

Simple meaning

Selected neurons learn from task error; the remaining neurons learn local activity patterns.

Activity controller and current result

Controller

$$q_j(t) = \frac{s_j(t) + \epsilon}{l_j(t) + \epsilon}$$

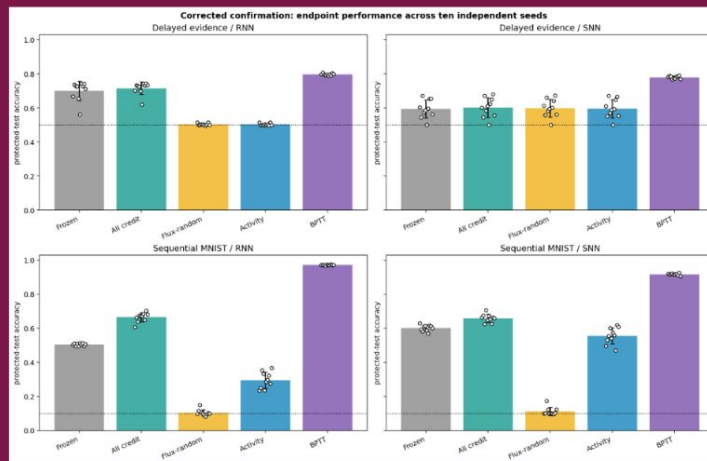
- $s_j(t)$: recent activity
- $l_j(t)$: long-term activity
- top- k neurons receive credit
- others receive Oja learning

Fair controls

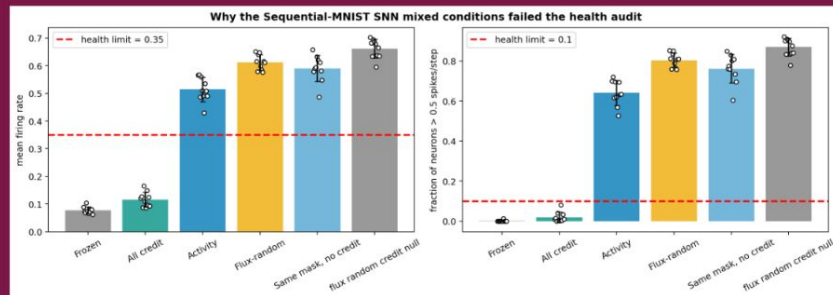
- same credit occupancy
- matched switching flux
- same mask with credit removed
- frozen recurrence and BPTT references

Conclusion

Local credit contains useful task information, but activity based allocation was not consistently better than matched controls. Mixed Sequential-MNIST SNNs also showed excessive firing.



Endpoint performance across ten seeds.

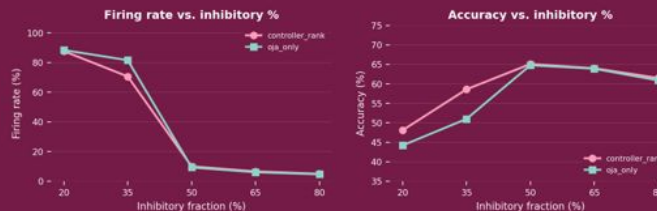
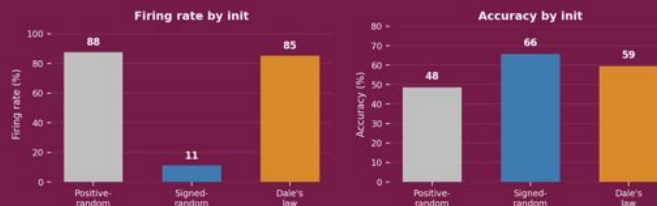
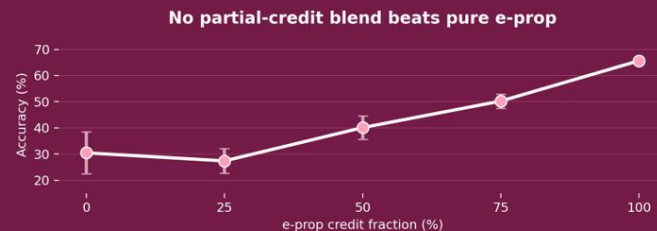
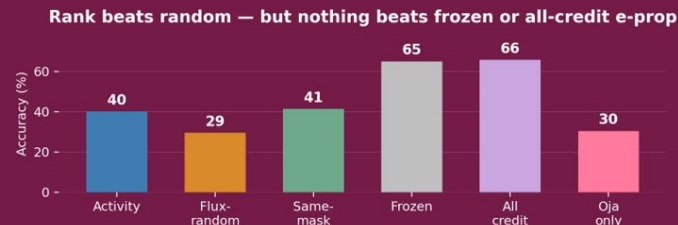


Mixed SNN conditions exceeded the firing-health limits.

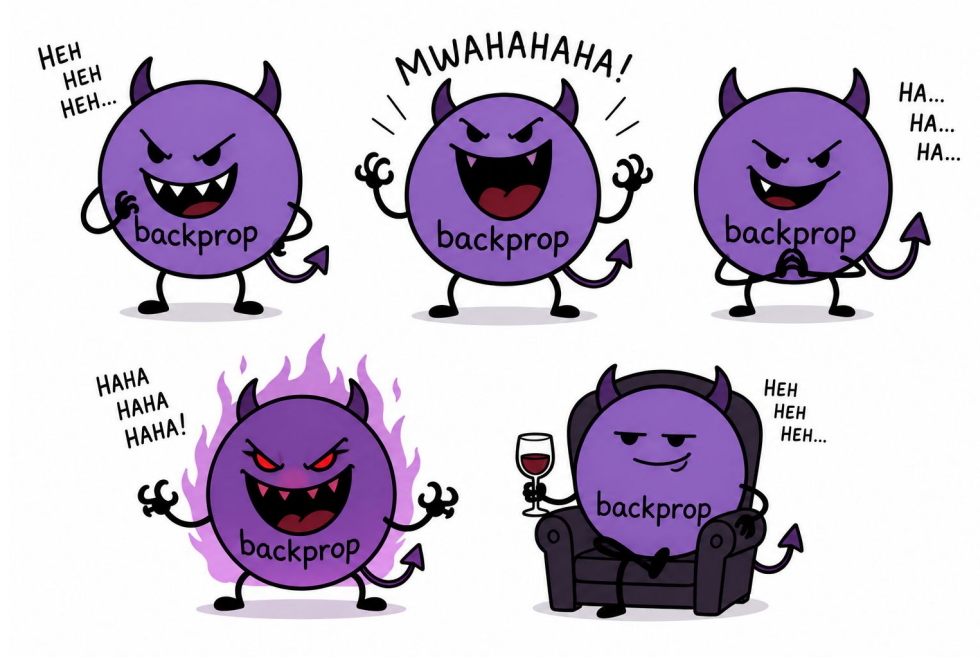
SNNs

Same controller as shown in the previous slide; Tested on neuromorphic MNIST (N-MNIST)

- Controllers - Top-50% by recent/long-term activity ratio vs probabilistically switching
- Activity-based switching beat random switching everytime, but neither beat frozen or all-credit e-prop (right)
- Oja's decay term was too weak, causing exploding weights; fixed with an explicit row-norm cap
- Recurrent weights started all-excitatory, causing saturated firing; fixed with signed-random init

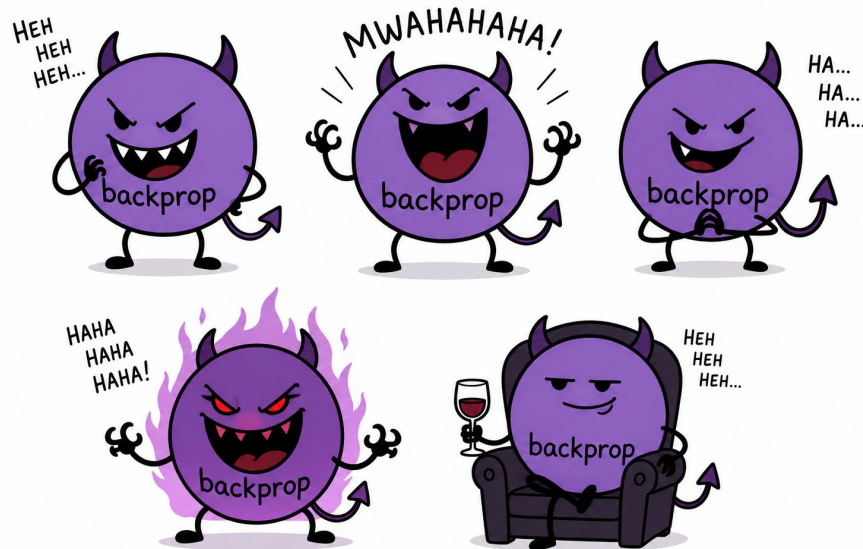


Conclusion and Future Directions



Conclusion and Future Directions

- The task needs to be designed so that the different learning rules can contribute. Otherwise, BPTT is the optimal update.
 - For SNNs: using Oja for pre-training followed by Eprop training, results in slightly higher accuracy than using Eprop throughout the entire process.
-
- Evaluate on biologically relevant datasets.
 - Investigate additional local learning rules
 - Investigate Hebbian meta-learning rate trial's non-convergence.



Thanketh Thee

